

01-27-2025

→ HW1 due Monday

→ Review:

1) $C_t = E_t C_{t+1}$ if $\beta(1+r)=1$, $U(C) = \alpha C - \frac{\gamma}{2} C^2$

2) $\sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t} = a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} \leftarrow BC$

$E_0 \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t} = a_0 + E_0 \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t}$ (not BC but exp of)

$\hookrightarrow E_0 C_0 + \frac{E_0 C_1}{(1+r)} + \frac{E_0 C_2}{(1+r)^2} + \dots = C_0 \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} = \frac{1+r}{r} C_0$
 $= \frac{1}{1-\beta} C_0$

$\hookrightarrow \frac{1+r}{r} C_0 = a_0 + E_0 \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} \Rightarrow C_0 = \frac{r}{1+r} \left(a_0 + E_0 \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} \right)$

$\hookrightarrow C_t = \frac{r}{1+r} \left(a_t(1+r) + E_t \sum_{j=0}^{\infty} \frac{y_{t+j}}{(1+r)^j} \right) = r a_t + \frac{r}{1+r} E_t \sum_{j=0}^{\infty} \frac{y_{t+j}}{(1+r)^j}$

→ $r a_t \Rightarrow$ MPS fin wealth

→ $\frac{r}{1+r} \Rightarrow$ MPS human wealth

$\hookrightarrow C_0 = (1-\beta) \left[a_0 + E_0 \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} \right]$

E [Permanent Income]

→ certainty Equivalence (CEq):

↳ as far as consumption is concerned, agent behaves as if future income is certain and equal to its expected value.

↳ Permanent Income Hyp. w/o uncertainty

$$C_0 = (1-\beta) \left[a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} \right]$$

↳ consumption is static despite changes in variance

→ 2-Period Example:

$$\{P_1, P_2\}; \{y_1, y_2\}; \{c_1, c_2\}$$

$$c_1 = y_1 + a_1 - a_2; c_2 = a_2(1+r) + y_2$$

s.t.

$$\beta(1+r) = 1; u(c) = \alpha c - \frac{\gamma}{2} c^2$$

↳ no uncertainty

$$U'(c_1) = U'(c_2) \Rightarrow c_1 = c_2$$

$$y_1 + a_1 - a_2 = a_2(1+r) + y_2$$

$$a_2 = y_1 - y_2 + a_1 \cdot (1+r)^{-1}$$

$$c_1 = y_1 + a_1 - \frac{y_1 - y_2 + a_1}{1+r} \Rightarrow (y_1 + a_1) \left(1 - \frac{1}{1+r} \right) + \frac{y_2}{1+r}$$

$$= y_1 + a_1 \frac{1+r}{1+r} + \frac{y_2}{1+r} \Rightarrow \frac{1+r}{1+r} \left[\underbrace{a_1}_{\text{f. wealth}} + \underbrace{y_1 + \frac{y_2}{1+r}}_{\text{human wealth}} \right]$$

→ 2-Period Example:

$$\{P_1, P_2\}; \{y_1, y_2\}; \{c_1, c_2\}$$

$$c_1 = y_1 + a_1 - a_2; c_2 = a_2(1+r) + y_2$$

s.t.

$$\beta(1+r) = 1; u(c) = \alpha c - \frac{\gamma}{2} c^2$$

↳ With uncertainty

$$u'(c_1) = E u'(c_2)$$

$$\alpha - \gamma c_1 = E[\alpha - \gamma c_2] \Rightarrow c_1 = E c_2$$

$$y_1 + a_1 - a_2 = a_2(1+r) E(y_2)$$

$$a_2 = \frac{a_1 + y_1 + E y_2}{2+r}$$

$$c_1 = \frac{1+r}{2+r} \left(a_1 + y_1 + \frac{E y_2}{1+r} \right)$$

→ PIH

↳ No Uncertainty

$$c_t = c_{t+1}$$
$$c_t = r a_t + \frac{r}{1+r} \sum_{j=0}^{\infty} \frac{y_{t+j}}{(1+r)^j}$$

↳ With:

$$c_t = E_t c_{t+1}$$
$$c_t = r a_t + \frac{r}{1+r} E_t \sum_{j=0}^{\infty} \frac{y_{t+j}}{(1+r)^j}$$

→ Innovations to PIH: P_t

$$\hookrightarrow P_{t+1} - E_t P_{t+1} = c_{t+1} - E_t c_{t+1}$$

$$P_{t+1} - E_t P_{t+1} = r a_{t+1} + \frac{1}{1+r} E_{t+1} \sum_{j=0}^{\infty} \frac{y_{t+1+j}}{(1+r)^j} - \left(r E_t a_{t+1} + \frac{r}{1+r} E_t E_{t+1} \sum_{j=0}^{\infty} \frac{y_{t+1+j}}{(1+r)^j} \right)$$

$$\Rightarrow \frac{r}{1+r} \sum_{j=0}^{\infty} \left[E_{t+1} y_{t+1+j} - E_t y_{t+1+j} \right] \frac{1}{(1+r)^j}$$

$$c_{t+1} - c_t = \frac{r}{1+r} \sum_{j=0}^{\infty} \frac{1}{(1+r)^j} (E_{t+1} y_{t+1+j} - E_t y_{t+1+j}) = \Delta c_{t+1}$$

↳ change due to new info

→ Specific Income Processes:

$$\hookrightarrow \text{AR}(1): y_{t+1} = \rho y_t + \varepsilon_t, \varepsilon_t \sim \text{i.i.d } N$$

$$j=0: E_{t+1} y_{t+1+j} - E_t y_{t+1+j} \Rightarrow E_{t+1} y_{t+1} - E_t y_{t+1}$$

$$= E_{t+1} (\rho y_t + \varepsilon_{t+1}) - E_t (\rho y_t + \varepsilon_{t+1})$$

$$= \rho y_t + \varepsilon_{t+1} - \rho y_t \Rightarrow \varepsilon_{t+1} \text{ (info gained at } t+1)$$

$$j=1: E_{t+1} y_{t+2} - E_t y_{t+2} \Rightarrow \rho^2 y_t + \varepsilon_{t+1} + \varepsilon_{t+2} = y_{t+2}$$